

(d) Collecting data		
1d.01	a. Know that data can be collected from different sources: experimental (laboratory, field and natural), simulation, questionnaires, observation, reference, census, population and sampling. b. Know that sources of secondary data should be acknowledged.	The design of data collection sheets is expected. Simulations may include use of random numbers.
1d.02	<u>Know the importance of reliability and validity with regard to collected data.</u>	<u>Reliability is the extent to which repeated measurements yield similar results.</u> <u>Validity is the extent to which a test measures what was intended.</u>

What students need to learn:		Guidance
(d) Collecting data <i>continued</i>		
1d.03	<u>Determine factors that may lead to bias, including issues of sensitivity of the content matter, and know how to minimise data distortion.</u>	
1d.04	Know the key features to be considered when planning data collection: leading questions, avoiding biased sources, time factors, open/closed questions, different types of interview technique.	The design of suitable questions and data collection sheets is expected. Awareness of the advantages and disadvantages of data collection techniques. The rationale behind pilots for questionnaires and pre-tests for experiments should be known.
1d.05	Know and demonstrate understanding of techniques used to deal with problems that may arise with collected data.	For example missing data, incorrect formats, non-responses, incomplete responses, etc.
1d.06	Know why data may need to be 'cleaned' before further processing, including issues that arise on spreadsheets and apply techniques to clean data in context.	In the pre-processing stage: consideration of genuine and other outliers and anomalies, or removal of extraneous symbols or notation when using technology (e.g. spreadsheets or statistical software). See also 2c.03.
1d.07	<u>Know the importance of identifying and controlling extraneous variables.</u>	
2 Processing, representing and analysing data		